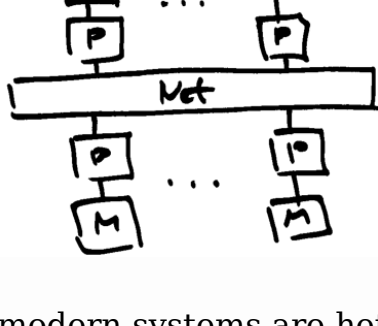


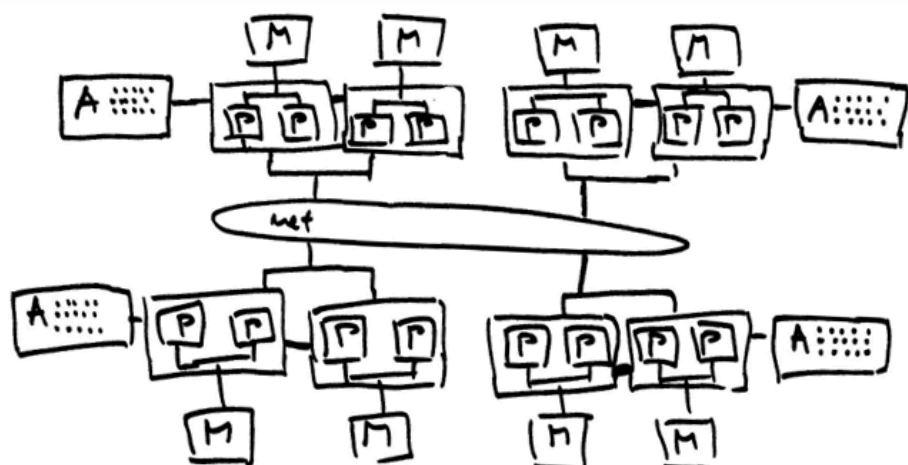
Distributed memory systems

Architecture

- multiple instructions, multiple data
- processors have their own memory
- other processors do not see memory changes
- processors exchange data by sending messages
- slower compared to the shared memory systems
- more scalable
- focus on interconnections
- off-the-shelf hardware components (cost effective)
- first systems had one core per node



- modern systems are heterogeneous
 - many nodes connected
 - shared memory within a node
 - accelerators on some nodes
 - message passing between nodes
 - programming of modern systems reflects architectural design



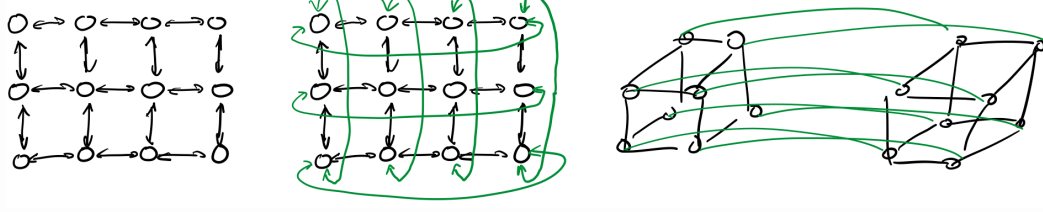
- network or interconnect is what distinguishes distributed systems
 - distributed memory parallel computers are just regular computers, nodes programmed like any other
 - designing and programming the distributed memory systems means thinking about how data moves between the compute nodes

Interconnect

- important characteristics
 - How are the nodes connected?
 - How are the nodes attached to the network?
 - What is the performance of the network?
- direct and indirect topologies
 - each node has a switch in direct topologies
 - there are more switches than nodes in indirect topologies

Direct topologies

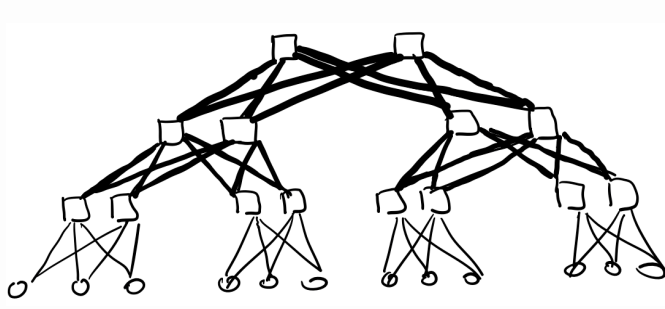
- mesh and torus
 - 1D, 2D, 3D, xD
 - torus links mesh ends together
 - hypercube



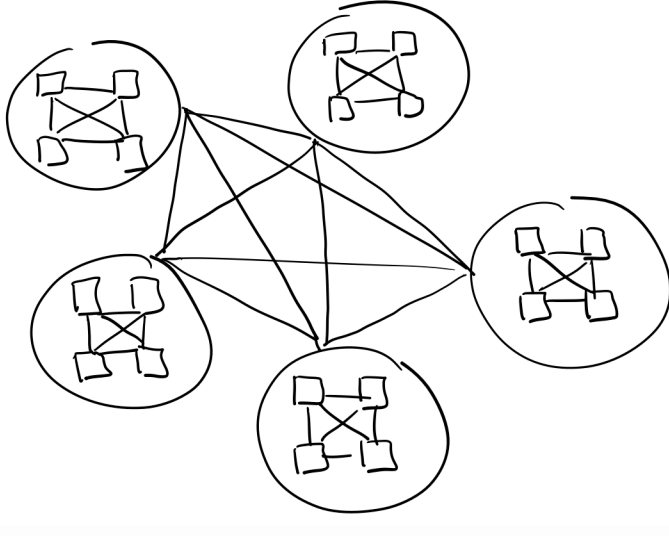
- communication is efficient among neighboring nodes
- constant cost to scale to more nodes
- simple routing algorithms
- easy to understand, simple to model
- matches many problems well

Indirect topologies

- multilevel networks
 - fat-tree network



- dragonfly

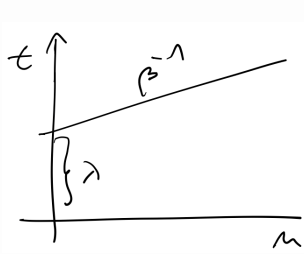


- better throughput than direct topologies
 - reduced number of hops
- cost grows faster than linear with additional compute nodes
- more complex, harder to model

Modelling throughput

- simple model
 - latency λ (seconds) needed to establish a connection
 - bandwidth β (bits per second)
 - time needed to transfer a message of length n (bytes, byte equals eight bits):

$$t(n) = \lambda + \beta^{-1} n$$



- improvements
 - account for the number of hops
 - additional network characteristics
 - topology, number of simultaneous connections, ...